A Real-Time Spectrum Analyser and FM Receiver Using an RTL-SDR and GNU Radio

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Report type	Technical report
Status	Implemented; live off-air reception and FM demodulation demonstrated
Tools	RTL-SDR (R820T2 tuner, 8-bit ADC), GNU Radio Companion, Python
Date	2026

Abstract. A spectrum analyser and wideband FM receiver were built from a low-cost RTL-SDR and GNU Radio. Complex baseband (I/Q) samples at 2.4 MS/s feed two paths: a windowed FFT producing live spectrum and waterfall displays, and a demodulation chain (frequency translation, low-pass filtering, decimation by 10, quadrature discrimination, de-emphasis) producing audio. Centre frequency, sample rate and RF gain are runtime controls. At 2.4 MS/s a 1024-point FFT gives a resolution bandwidth of 2.34 kHz over a 2.4 MHz span. Broadcast FM at 98.4 MHz was received and demodulated. The 8-bit ADC limits ideal quantisation SNR to about 49.9 dB, and the FFT adds about 27 dB of processing gain. Observations of noise floor and dynamic range are qualitative; no calibrated power measurements were made.

Index terms: software-defined radio, RTL-SDR, complex baseband, FFT, resolution bandwidth, windowing, FM demodulation.

1. Objective

The objective was to build a receive chain from complex samples to audio and to relate its configuration (sample rate, FFT length, window, filtering) to what the instrument can resolve.

2. System

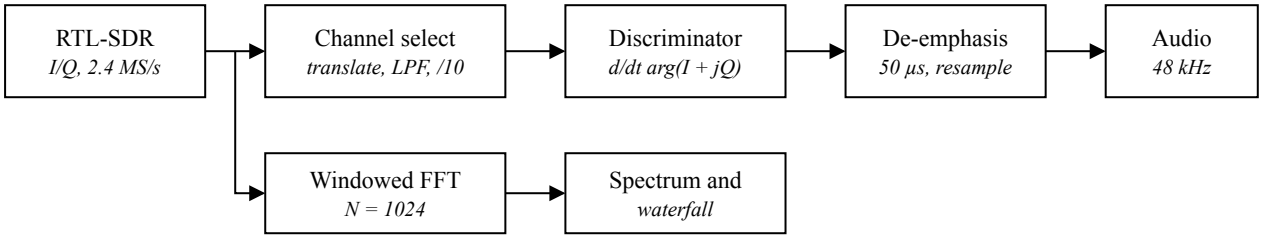


Figure 1. Receive chain. The same I/Q stream feeds the demodulation path (top) and the spectral display path (bottom).

The RTL-SDR uses a zero-IF tuner followed by an 8-bit ADC, producing I and Q channels that form the complex signal I + jQ. Because positive and negative frequencies are distinguishable, the usable span equals the full sample rate f_s ($\pm f_s/2$ around the tuned centre).

3. Spectral Analysis

$$RBW = f_s / N_{FFT} \tag{1}$$

Table 1. Resolution bandwidth against FFT length (calculated).

N_{FFT}	RBW at 2.4 MS/s	Frame duration
1024 (used)	2.34 kHz	0.43 ms
2048	1.17 kHz	0.85 ms
4096	0.59 kHz	1.71 ms

Longer transforms resolve closer signals at the cost of time resolution. Each block is windowed before the transform to reduce spectral leakage; the window trades main-lobe width against side-lobe level (Table 2). The waterfall adds a time axis that shows intermittent and drifting signals.

Table 2. Window properties (from [4]).

Window	Highest side lobe	ENBW (bins)	Scalloping loss
Rectangular	−13 dB	1.00	3.9 dB
Hann	−31 dB	1.50	1.4 dB
Blackman-Harris (4-term)	−92 dB	2.00	0.8 dB

4. FM Demodulation

1. **Channel selection.** The wanted carrier is translated to 0 Hz, low-pass filtered (100 kHz cut-off, 20 kHz transition) and decimated by 10 to 240 kS/s.
2. **Discrimination.** The message is the instantaneous frequency, recovered as d/dt arg(I + jQ). Broadcast FM uses a peak deviation of 75 kHz.
3. **Audio.** A 50 μs de-emphasis filter (UK and Europe) reverses transmitter pre-emphasis, and the signal is resampled to 48 kHz.

$$B \approx 2(\Delta f + f_m) = 2(75 + 15)\ \text{kHz} = 180\ \text{kHz} \tag{2}$$

Carson's rule (2) gives an occupied bandwidth of about 180 kHz, consistent with the approximately 200 kHz channel filter.

5. Dynamic Range

$$SNR_{ADC} = 6.02N + 1.76\ \text{dB}, \quad G_{FFT} = 10\log_{10}(N_{FFT}/2) \tag{3}$$

For N = 8 bits, $SNR_{ADC} \approx 49.9$ dB; a 1024-point FFT adds $G_{FFT} \approx 27$ dB, so narrowband signals below the broadband quantisation floor can appear on the averaged spectrum. Averaging more FFT frames lowers the variance of the displayed noise floor at the cost of update rate. With strong and weak signals in the band, ADC spurs became visible, illustrating the dynamic-range limit of the front end (qualitative observation).

6. Configuration

Table 3. Receiver configuration used for broadcast FM.

Parameter	Value
Receiver	RTL-SDR, R820T2 tuner, 8-bit ADC
Sample rate	2.4 MS/s complex
Centre frequency	98.4 MHz (retunable across HF, VHF and UHF)
RF gain	40 dB
FFT size	1024 (RBW 2.34 kHz)
Channel filter	Low-pass 100 kHz; decimation 10
Demodulator	Wideband FM, quadrature discriminator
De-emphasis	50 µs
Audio output	48 kHz

7. Limitations

- The power axis is uncalibrated, so noise floor, SNR and occupied bandwidth were not measured quantitatively.
- Tuner frequency error, I/Q imbalance and DC offset were not characterised or corrected.
- The effect of sample rate, FFT size and gain was analysed but not measured systematically.
- Performance depends on the antenna used, which is not characterised.

8. Conclusion and Future Work

The receiver produced live spectral displays and demodulated broadcast FM, and its resolution and dynamic-range limits are consistent with the calculated values. Future work: calibrate the power axis against a signal generator; measure noise floor and SNR against RF gain and FFT averaging; calibrate tuner frequency error against a known carrier; correct I/Q imbalance; and add stereo decoding (19 kHz pilot, 38 kHz L−R subcarrier) and RDS.

Appendix A. Flowgraph

```
from gnuradio import gr, filter, analog, audio
import osmosdr

samp_rate, center, audio_rate, decim = 2.4e6, 98.4e6, 48000, 10

class fm_rx(gr.top_block):
    def __init__(self):
        gr.top_block.__init__(self)
        self.src = osmosdr.source(args="rtl=0")
        self.src.set_sample_rate(samp_rate)
        self.src.set_center_freq(center)
        self.src.set_gain(40)
        taps = filter.firdes.low_pass(1.0, samp_rate, 100e3, 20e3)
        self.chan = filter.freq_xlating_fir_filter_ccc(decim, taps, 0, samp_rate)
        self.demod = analog.wfm_rcv(quad_rate=samp_rate/decim,
                                    audio_decimation=int((samp_rate/decim)/audio_rate))
        self.sink = audio.sink(audio_rate)
        self.connect(self.src, self.chan, self.demod, self.sink)

if __name__ == "__main__":
    fm_rx().run()
```

Listing A1. Wideband FM receiver. A parallel FFT sink (not shown) drives the spectrum and waterfall.

References

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[5] The GNU Radio Project, "GNU Radio Companion," documentation, gnuradio.org.